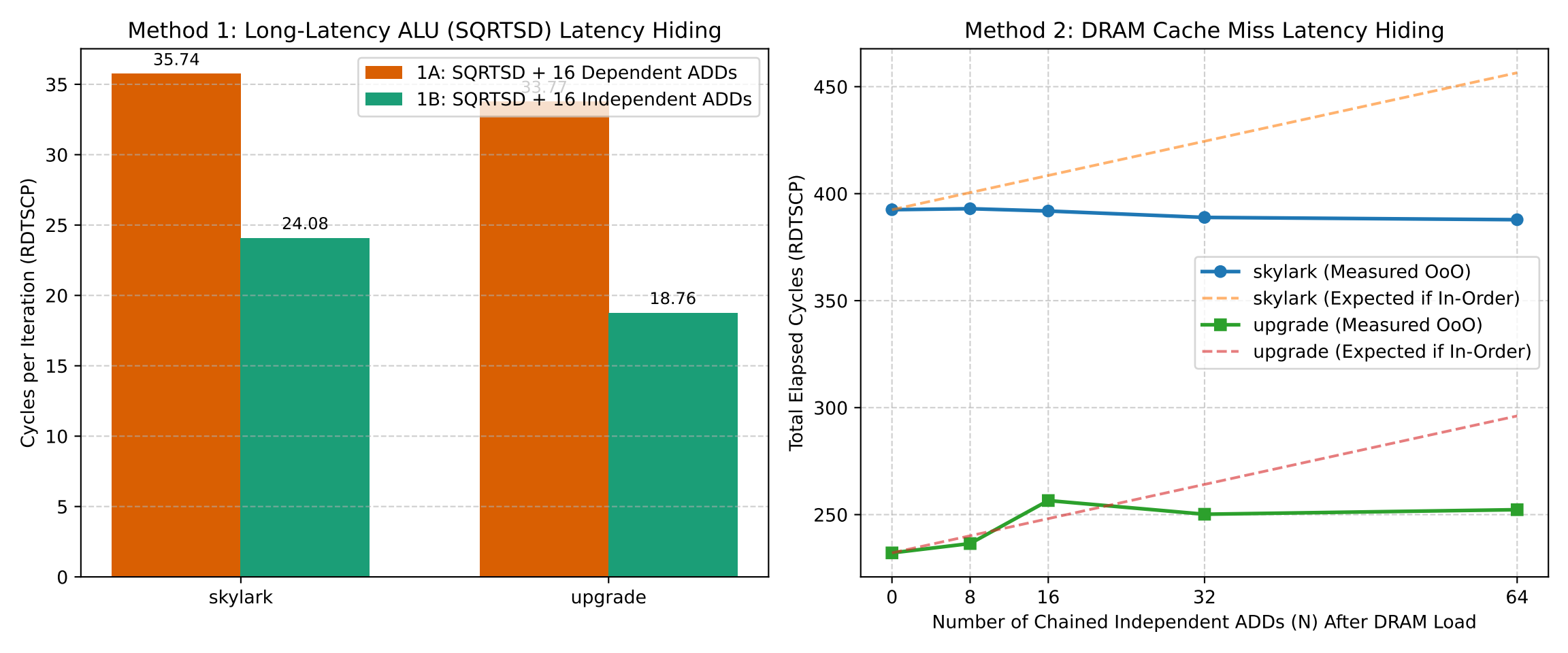
Method 1: Long-Latency ALU (SQRTSD) Latency Hiding
1A: SQRTSD + 16 Dependent ADDs
1B: SQRTSD + 16 Independent ADDs
35.74
24.08
33.77
18.76
Cycles per Iteration (RDTSCP)
0
5
10
15
20
25
30
35
skylark
upgrade
Method 2: DRAM Cache Miss Latency Hiding
skylark (Measured OoO)
skylark (Expected if In-Order)
upgrade (Measured OoO)
upgrade (Expected if In-Order)
Total Elapsed Cycles (RDTSCP)
250
300
350
400
450
0
8
16
32
64
Number of Chained Independent ADDs (N) After DRAM Load